V.L Memorial Public School

475 Vijay Nagar

CHAPTER-10-QUESTIONS-CLASS8TH-MATHS

Class 08 - Maths (Ganita Prakash)

Time Allowed: 6 hours **Maximum Marks: 158**

Section A

1. **Fill in the blanks:** **[6]**
   (a) A recipe requires 2 cups of flour for 8 cookies. For 32 cookies, _____ cups of flour are needed. **[1]**
   (b) In a map, if RF is 1 : 1,00,000, then 3 cm on the map represents _____ km on the ground. **[1]**
   (c) A quantity of 240 is divided in the ratio 2 : 3. The first part is _____. **[1]**
   (d) RF stands for _____ Fraction. **[1]**
   (e) If 8 workers can complete a job in 12 days, then 4 workers will take _____ days. **[1]**
   (f) If 4 : 6 :: 10 : x, then the value of x is _____. **[1]**

Section B

2. A car travels 150 km in 3 hours. At the same speed, how far will it travel in 7 hours? **[1]**

   a) 300 km b) 400 km

   c) 350 km d) 250 km

3. The ratio of apples to oranges in a basket is 7 : 3. If there are 50 fruits in total, how many oranges are there? **[1]**

   a) 30 b) 25

   c) 20 d) 15

4. If 12 notebooks cost ₹ 360, how much will 20 notebooks cost at the same rate? **[1]**

   a) ₹ 720 b) ₹ 500

   c) ₹ 600 d) ₹ 480

5. Which of the following pairs of ratios are proportional? **[1]**

   a) 3 : 4 and 8 : 5 b) 5 : 6 and 10 : 12

   c) 2 : 3 and 5 : 9 d) 4 : 7 and 12 : 20

6. If 5 kg of rice costs ₹ 250, what is the cost of 8 kg at the same rate? **[1]**

   a) ₹ 400 b) ₹ 450

   c) ₹ 500 d) ₹ 350

7. Which of the following vary inversely with each other? **[1]**

   a) Speed and distance covered. b) Distance travelled and time taken.

   c) Distance covered and taxi fare. d) Speed and time taken.

8. If 3 liters of paint cover $12\ m^2$ of wall, how many liters are needed to cover $30\ m^2$? **[1]**

a) 8 b) 6

c) 7.5 d) 9

9. A recipe uses 4 cups of sugar to make 20 sweets. How many cups of sugar are needed to make 50 sweets? **[1]**

a) 6 b) 8

c) 12 d) 10

10. The ratio of three numbers is 5 : 7 : 9. **[1]**
Which of the following could be their actual values?

a) 10, 14, 18 b) 20, 28, 30

c) 15, 21, 24 d) 25, 30, 45

11. Which of the following best describes the meaning of RF = 1 : 50,000 ? **[1]**

a) 50,000 cm on map = 1 cm on ground b) 1 km on map = 50,000 km on ground

c) 1 cm on map = 50,000 cm on ground d) 50,000 km on map = 1 km on ground

12. Which pair of quantities is in inverse proportion? **[1]**

a) cost and number of items at fixed price b) speed and time for fixed distance

c) distance and time at constant speed d) speed and distance for fixed time

13. A quantity is divided in the ratio 1 : 2 : 3. **[1]**
Which expression represents the third part of x?

a) x / 6 b) 3x / 6

c) x / 3 d) 2x / 6

14. Divide 360 in the ratio 2 : 4 : 6. **[1]**
The middle part is

a) 180 b) 120

c) 60 d) 90

15. A pump fills a tank in 6 hours. **[1]**
How long will 3 identical pumps take to fill the same tank?

a) 3 hours b) 4 hours

c) 2 hours d) 1 hour

16. If a : b : c = 2 : 3 : 4 and a = 10, then b equals **[1]**

a) 20 b) 15

c) 12 d) 18

17. In a class, the number of students choosing three sports is in the ratio 2 : 3 : 5. **[1]**
If the pie chart is drawn, the angle for the second sport is

a) 120° b) 108°

c) 90°

d) 72°

18. If 4 : 6 : 10 :: p : 9 : q, then the values of p and q are **[1]**

a) p = 6, q = 15

b) p = 8, q = 20

c) p = 6, q = 10

d) p = 12, q = 30

19. A map has a scale of 1 : 50,00,000. If two cities are 3 cm apart on the map, what is their actual geographical distance? **[1]**

a) **1.5 km**

b) **150 km**

c) **1500 km**

d) **15 km**

20. If the ratios 7 : x and 21 : 12 are proportional, then x equals **[1]**

a) 6

b) 5

c) 4

d) 3

21. A pie chart shows 54° for one category. **[1]**
What fraction of the whole does it represent?

a) 3/10

b) 1/5

c) 1/8

d) 3/20

Section C

22. **Assertion (A):** The ratios 6 : 8 and 9 : 12 are proportional. **[1]**
Reason (R): Two ratios are proportional if the sum of their terms is the same.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

23. **Assertion (A):** If the number of workers increases, the number of days required to complete a fixed job also increases. **[1]**
Reason (R): For a fixed amount of work, number of workers and number of days are inversely proportional.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** If the RF of a map is 1 : 50,000, then 2 cm on the map represents 1 km on the ground. **[1]**
Reason (R): In RF, actual distance is found by multiplying the map distance by the scale number.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** If 300 is divided in the ratio 2 : 3, then the first part is 120. **[1]**
Reason (R): To divide a quantity in a ratio, we divide the quantity by the difference of the parts.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** The ratios 3 : 6 : 9 and 1 : 2 : 3 are proportional. **[1]**

Reason (R): Corresponding terms of proportional ratios have equal differences

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** If 10 men can make a wall in 15 days, then 3 men can make the wall in 45 days. **[1]**

Reason (R): Two quantities x and y are said to be in 'direct proportion' if they increase (or decrease) together in such a manner that the ratio of their corresponding values remains constant.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. **Assertion (A):** If an amount of fodder lasts for 45 days for 124 goats, then for 52 days it will last for 90 goats at the same rate. **[1]**

Reason (R): The number of goats and the number of days fodder lasts show an inverse relation between the two quantities.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

Section D

29. A car takes 3 hours to travel a fixed distance at a speed of 40 km/h. **[1]**

How long will it take at 60 km/h?

30. Divide 360 in the ratio 1 : 2 : 3. **[1]**

31. If 5 workers can complete a job in 12 days, how many days will 10 workers take? **[1]**

32. Write the simplest form of the ratio 6 : 9 : 12. **[1]**

33. If x varies directly as y^2 and $x = 4$ when $y = 5$, then find x when y is 15. **[1]**

[Hint: $\frac{x}{y^2} = \frac{4}{5^2}$

$\Rightarrow \frac{x}{y^2} = \frac{4}{25}$

$\Rightarrow 25x = 4y^2$

Now put $y = 15$ and find x]

34. The RF of a map is 1 : 2,00,000. **[1]**

Find the ground distance represented by 4 cm on the map.

35. Check whether the ratios 7 : 9 and 21 : 27 are proportional. **[1]**

36. The angles of a triangle are in the ratio 2 : 3 : 4. Find all the three angles and also find the greatest angle. **[2]**

37. If 4 notebooks cost ₹60, find the cost of 10 notebooks and also find the unit cost of one notebook. **[2]**

38. Kanika typed 1,440 words in 1 hour and 20 minutes. How many words would she type in 42 **[2]**

minutes?

39. Find the value of x in 3 : 5 :: x : 25 and also check whether 6 : 10 and x : 25 are proportional. **[2]**

40. An aeroplane flies 1600 km in 2 hours. At the same rate, how long will it take to fly 6400 km? **[2]**

41. Divide ₹ 540 in the ratio 2 : 3 : 4. **[2]**

42. In a pie chart, four sectors are in the ratio 1 : 2 : 3 : 4. Find the angle of the third sector and also find the angle of the smallest sector. **[2]**

43. One gas cylinder is used to cook food for a family of 8 people for 30 days. If 4 guests join the family, how long will the cylinder last? **[2]**

44. If 6 workers can complete a work in 15 days, how many days will 10 workers take and also find the number of workers required to complete the same work in 6 days. **[2]**

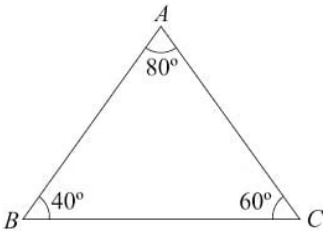
45. A distance of 2 cm on a map represents 1 km on the ground. Find the RF and also find the ground distance represented by 7 cm. **[2]**

46. Five tractors can mow a field in 8 days. If there are only 4 tractors available, in how many days will the field be mowed? **[3]**

47. A school has provisions for 60 students for 20 days. After 5 days, 20 new students join. For how many more days will the provisions last? **[3]**

48. In inverse proportion, if x increases from 20 to 50, by what factor does y change? If y was 15, what is the new value? **[3]**

49. A triangle has angles in ratio 2 : 3 : 4. A student constructs it with angles 40°, 60°, and 80°. Is this triangle correct? **[3]**



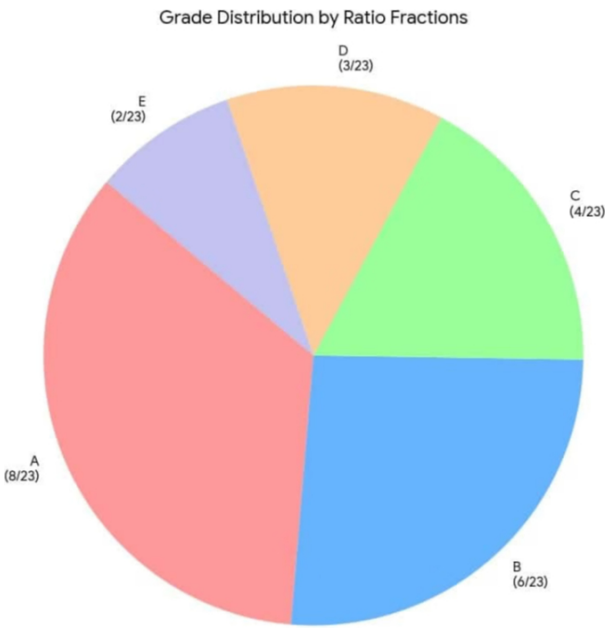
50. A recipe requires mixing ingredients in the ratio Flour : Sugar : Butter :: 5 : 2 : 3. If Meera has 15 cups of flour, how much sugar and butter should she use? What is the total quantity of the mixture? **[3]**

51. A family of 8 people has enough stock of rice to last for 30 days. Due to the arrival of some relatives, this food was consumed in 20 days. How many guests joined the family? **[3]**

52. A spice mix has ratio coriander : cumin : pepper :: 6 : 4 : 2. If you use 1 spoon of pepper, can you make the same mix using 2 spoons of coriander? Justify. **[3]**

53. In a survey of 720 students, the ratio of those who like Math, Science, and English is 4 : 3 : 5. Draw a table showing how many students like each subject. **[3]**

54. In a pie chart showing 5 grades with ratio A : B : C : D : E :: 8 : 6 : 4 : 3 : 2, what is the angle for grade C? A student says it's 62.6°. Verify this. **[3]**



55. A contractor undertook a contract to complete a part of a stadium in 9 months with a team of 560 workers. Later on, it was required to complete the job in 5 months. How many extra workers should he employ to complete the work? **[3]**

56. If a boat travels 125 km in $2\frac{1}{2}$ hours, how far will it travel in 12 hours? **[3]**

57. The angles of a triangle are in the ratio 2 : 3 : 4. Find all three angles of the triangle. **[3]**

58. Puneet has only 2 red chillies in his kitchen. But he wants to make spice mix powder that tastes the same as Viswanath's spice mix powder. How much of the other ingredients should Puneet use to make his spice mix powder? **[5]**

59. A school has 450 students. The ratio of students in Grade 6 : Grade 7 : Grade 8 is 4 : 5 : 6. How many students are there in each grade? **[5]**

60. Ram can paint a wall in 5 hours and Shyam can paint the same wall in 10 hours. If they work together, how long will it take them to paint the wall? **[5]**

61. A tank can be filled by 3 pipes in 6 hours. How long will it take to fill the tank if 2 more pipes of the same capacity are added? **[5]**

62. A school organized a survey among 720 students about their favourite sport. The results showed that the number of students who liked Cricket, Football, Badminton, and Others were in the ratio 5 : 4 : 3 : 2. Create a pie chart for this data and find how many students liked each sport. **[5]**

63. How do you construct a triangle with angles in a given ratio? Construct a triangle with angles in the ratio 1 : 3 : 5. **[5]**

64. A group of 720 people were asked to vote for their favourite fruit from three options: Mango, Apple, and Orange. 180 people liked Mango, 240 people liked Apple, and the rest liked Orange. Find the angle of each sector in a pie chart representing this data. **[5]**

65. A survey was conducted among 180 students to find their favourite sport. The results are as follows: **[5]**
    - Cricket: 60 students
    - Football: 45 students
    - Badminton: 30 students
    - Tennis: 27 students
    - Others: 18 students

Find the ratio of students preferring each sport in simplest form.

Calculate the angle for each sport in a pie chart.

Draw a rough sketch of the pie chart with labeled angles.

66. What is a proportional relationship? Explain with the example of making idli batter and verify if two given ratios are proportional. **[5]**

67. What is a Representative Fraction (RF) on a map? If a map has RF 1 : 60,00,000, what does it mean and how do you use it to find actual distances? **[5]**

Section E

68. **Read the following text carefully and answer the questions that follow:** **[4]**

A construction company needs to lay a road. They observe that 20 workers can complete the work in 4 days. The project manager wants to know how many days it will take if they reduce the number of workers to 10. He recalls that the number of workers and the number of days to complete a fixed amount of work are in inverse proportion. This means that if one quantity changes by a factor, the other changes by its reciprocal. Later, they need to calculate how many workers are needed to complete the work in just 2 days. Understanding inverse proportion helps in efficient workforce planning.

Questions:

i. What is the constant product of workers and days in this situation? **(1)**

ii. If the number of workers is halved, what happens to the number of days required? **(1)**

iii. How many days will 10 workers take to complete the same road? **(2)**

OR

If the work needs to be completed in 2 days, how many workers are required? **(2)**

69. **Read the following text carefully and answer the questions that follow:** **[4]**

Raj's father drives from their village to the nearby town every week. He notices that when he drives at 40 km/h, the journey takes 3 hours. He wants to understand the relationship between speed and time. He realizes that for a fixed distance, speed and time are in inverse proportion – if he increases his speed, the time decreases by the same factor. He plans to drive at 60 km/h next time. Later, he needs to reach the town in 2 hours for an important meeting and wants to know what speed he must maintain to achieve this.

Questions:

i. What is the constant distance between the village and the town? **(1)**

ii. If Raj's father increases his speed from 40 km/h to 60 km/h, by what factor does the speed increase? **(1)**

iii. How long will the journey take if he drives at 60 km/h? **(2)**

OR

If he needs to reach the town in 2 hours, at what speed must he drive? **(2)**

70. **Read the following text carefully and answer the questions that follow:** **[4]**

A residential school has food provisions to feed 80 students for 15 days. The school administrator receives news that 20 more students will be joining the school immediately. She needs to determine how long the existing food provisions will last with the increased number of students. She recognizes that the number of students and the number of days the food lasts are inversely proportional - more students means the food will last fewer days. Later, she wonders how many

students can be fed if the provisions need to last for 20 days instead of 15. This type of reasoning is essential for resource management.

Questions:

i. What is the total number of student-days of food available? **(1)**

ii. After 20 more students join, what is the new total number of students? **(1)**

iii. For how many days will the food provisions last after the new students join? **(2)**

OR

If the provisions need to last for 20 days, how many students can be fed? **(2)**